West Bengal Council of Higher Secondary Education

EXPLAINED ANSWER PAPER

(Paper 1A — Questions with Answers)

Modern Computer Application (COMA)

Class XII – Foundation of AI

Full Marks : 70

Detailed answer with explanation for each question

In this answer paper every question is restated and followed by a detailed model answer with explanation. Examples, code and comparison tables are included so that students can understand each topic deeply.

GROUP – A (Multiple Choice Type Questions) [21 × 1 = 21]

Q1. Artificial Intelligence is the science of making machines that can — (a) only store data (b) simulate human intelligence (c) only print (d) only transmit signals [1]

Answer & Explanation:

Correct answer: (b) **simulate human intelligence**.

AI is the branch of computer science that builds machines or software able to perform human intelligence-based tasks such as learning, reasoning, problem solving and decision making. Storing data (a) is a database job, printing (c) is a printer job and transmitting signals (d) is networking — none of these is intelligence. Example: Google Assistant understands a question and answers it.

Q2. The term "Artificial Intelligence" was first coined in the year — (a) 1943 (b) 1956 (c) 1969 (d) 1997 [1]

Answer & Explanation:

Correct answer: (b) **1956**.

In 1956, at the Dartmouth Conference, John McCarthy first used the term "Artificial Intelligence" — so this year is treated as the birth of AI. The 1943 idea of neural networks predates the name; 1969 and 1997 relate to later events (e.g., Deep Blue beat Kasparov in 1997).

Q3. Who is regarded as the "Father of AI"? (a) Charles Babbage (b) John McCarthy (c) Alan Turing (d) Dennis Ritchie [1]

Answer & Explanation:

Correct answer: (b) **John McCarthy**.

John McCarthy coined the term “Artificial Intelligence” and created the LISP language, so he is the Father of AI. Charles Babbage is the father of the mechanical computer, Alan Turing is the father of computing and the Turing Test, and Dennis Ritchie created the C language.

Q4. The test used to judge whether a machine can behave intelligently is the — (a) Litmus Test (b) Turing Test (c) IQ Test (d) Boolean Test [1]

Answer & Explanation:

Correct answer: (b) **Turing Test**.

Proposed by Alan Turing in 1950, in the Turing Test a human converses with both a machine and a human; if the human cannot tell which is the machine, the machine is judged intelligent. The Litmus Test is chemistry, the IQ Test measures human intelligence, and there is no “Boolean Test”.

Q5. Learning from labelled data is called — (a) Unsupervised learning (b) Supervised learning (c) Reinforcement learning (d) Deep sleep learning [1]

Answer & Explanation:

Correct answer: (b) **Supervised learning**.

In supervised learning the input comes with the correct output (label) and the model learns that relationship (e.g., spam/not-spam). Unsupervised learning has no labels, reinforcement learning learns through reward and penalty, and “deep sleep learning” does not exist.

Q6. Learning through rewards and penalties is — (a) Supervised (b) Unsupervised (c) Reinforcement (d) Semi-supervised [1]

Answer & Explanation:

Correct answer: (c) **Reinforcement learning**.

In reinforcement learning an agent interacts with an environment, earning a reward for correct actions and a penalty for wrong ones, and learns the best policy by trial and error. Examples: self-driving cars, chess-playing AI. Supervised/unsupervised depend on the presence or absence of labels, not on rewards.

Q7. A neural network with many hidden layers is the basis of — (a) Data mining (b) Deep learning (c) OLAP (d) DBMS [1]

Answer & Explanation:

Correct answer: (b) **Deep learning**.

Deep learning is a branch of machine learning that uses deep neural networks with many hidden layers to learn complex patterns (e.g., face recognition, speech recognition). Data mining finds patterns in data, OLAP is analytical processing, and DBMS manages data.

Q8. NLP stands for — (a) Natural Language Processing (b) Neural Logic Programming (c) Network Layer Protocol (d) Non-Linear Programming [1]

Answer & Explanation:

Correct answer: (a) **Natural Language Processing**.

NLP is the branch of AI that enables a computer to understand, analyse and generate human language (English, Bengali, etc.). Examples: Google Translate, chatbots, voice assistants. The other options are not the real full form.

Q9. Enabling a computer to "see" and interpret images is called — (a) Computer Vision (b) Computer Graphics (c) Cloud Computing (d) Data Warehousing [1]

Answer & Explanation:

Correct answer: (a) **Computer Vision**.

Computer Vision extracts meaningful information from images or video (e.g., face recognition, reading number plates). Computer Graphics creates images (the reverse task), Cloud Computing provides services over the internet, and Data Warehousing stores data.

Q10. Detecting unusual credit-card transactions is an example of AI in — (a) Education (b) Fraud detection (c) Gaming (d) Weather [1]

Answer & Explanation:

Correct answer: (b) **Fraud detection**.

AI learns the pattern of normal transactions and flags abnormal (fraudulent) ones — an important application of AI in finance. Education, gaming and weather are unrelated to this task.

Q11. Which of the following is NOT a type of Machine Learning? (a) Supervised (b) Unsupervised (c) Reinforcement (d) Productive [1]

Answer & Explanation:

Correct answer: (d) **Productive**.

The three standard types of machine learning are supervised, unsupervised and reinforcement learning. "Productive learning" is not a recognised ML category.

Q12. AlphaGo or a chess-playing AI uses which type of learning? (a) Supervised (b) Reinforcement (c) Unsupervised (d) Batch [1]

Answer & Explanation:

Correct answer: (b) **Reinforcement**.

Game-playing AI like AlphaGo learns the best moves through trial and error, receiving rewards for winning and penalties for losing — the hallmark of reinforcement learning.

Q13. Clustering is an example of which type of learning? (a) Supervised (b) Unsupervised (c) Reinforcement (d) Labelled [1]

Answer & Explanation:

Correct answer: (b) **Unsupervised**.

Clustering groups similar data points without any labels, discovering hidden structure — a classic unsupervised learning task (e.g., customer segmentation).

Q14. A chatbot is mainly used in which field of AI? (a) Agriculture (b)

Customer service (c) Mining (d) Weather **[1]**

Answer & Explanation:

Correct answer: (b) **Customer service**.

Chatbots automatically converse with customers to answer queries 24×7, widely used in customer service, online shopping and banking helpdesks.

Q15. Deep learning is inspired by the structure of — (a) the human brain (neural network) (b) the hard disk (c) the keyboard (d) the printer [1]

Answer & Explanation:

Correct answer: (a) **the human brain (neural network)**.

Deep learning uses artificial neural networks modelled on the interconnected neurons of the human brain. Hard disk, keyboard and printer are hardware unrelated to its design.

Q16. The main cause of bias in an AI system is — (a) a fast processor (b) biased training data (c) a large monitor (d) more RAM [1]

Answer & Explanation:

Correct answer: (b) **biased training data**.

If an AI model is trained on biased or unbalanced data, it produces biased, unfair decisions. Processor speed, monitor size and RAM are hardware factors that do not cause bias.

Q17. A self-driving car identifies roads and pedestrians using — (a) Computer Vision (b) DBMS (c) OLAP (d) HTML [1]

Answer & Explanation:

Correct answer: (a) **Computer Vision**.

A self-driving car uses computer vision to "see" and interpret the road, signs and pedestrians from camera input. DBMS, OLAP and HTML have no role in visual perception.

Q18. Algorithmic trading is an application of AI in which field? (a) Education (b) Finance (c) Entertainment (d) Transport [1]

Answer & Explanation:

Correct answer: (b) **Finance**.

Algorithmic trading uses AI to analyse markets and buy/sell shares automatically at the right time — an application of AI in finance.

Q19. Personalized learning and intelligent tutoring are applications of AI in — (a) Finance (b) Education (c) Agriculture (d) Sports [1]

Answer & Explanation:

Correct answer: (b) **Education**.

Personalized learning and intelligent tutoring systems adapt lessons to each student's pace and needs — applications of AI in education (e.g., Duolingo, Khan Academy).

Q20. Which of the following is an ethical concern of AI? (a) Bias and dis-

crimination (b) Fast speed (c) Low cost (d) Good graphics **[1]**

Answer & Explanation:

Correct answer: (a) **Bias and discrimination**.

A model trained on biased data can make discriminatory decisions — a major ethical concern. Fast speed, low cost and good graphics are technical benefits, not ethical issues.

Q21. Google Translate is an example of — (a) NLP (b) OLAP (c) DBMS (d) DDL **[1]**

Answer & Explanation:

Correct answer: (a) **NLP**.

Google Translate understands text in one language and generates it in another — a real-world application of Natural Language Processing. OLAP, DBMS and DDL are database concepts.

GROUP – B (Short Answer Type Questions) [14 × 1 = 14]

Q1. Define Artificial Intelligence. [2]

Answer & Explanation:

Artificial Intelligence (AI) is a branch of computer science concerned with building intelligent machines or software that can perform human intelligence-based tasks such as learning, reasoning, understanding language, problem solving and decision making. Examples: ChatGPT, self-driving cars, voice assistants.

Q2. State any two important milestones in the history of AI. [2]

Answer & Explanation:

(1) 1950 — Alan Turing proposed the "Turing Test". (2) 1956 — John McCarthy coined the term "Artificial Intelligence" at the Dartmouth Conference. (Extra: in 1997 IBM's Deep Blue defeated world champion Garry Kasparov at chess.)

Q3. Write any two differences between human intelligence and artificial intelligence. [2]

Answer & Explanation:

(1) Human intelligence is natural and experience-based, whereas AI is man-made and data-driven. (2) Humans use emotion, creativity and common sense; AI only performs specific tasks fast and tirelessly but does not understand emotion.

Q4. What is Machine Learning? [2]

Answer & Explanation:

Machine Learning (ML) is a branch of AI in which a computer learns and improves from data on its own, instead of being explicitly programmed with every rule. The more data, the better the result. Example: an email spam filter.

Q5. Distinguish between supervised and unsupervised learning. [2]

Answer & Explanation:

Supervised learning trains a model on labelled data (input + correct output), e.g., recognising "cat/dog" in images. Unsupervised learning has no labels; the model finds groups or patterns (clusters) in the data on its own, e.g., grouping similar customers.

Q6. What is reinforcement learning? Give one example. [2]

Answer & Explanation:

In reinforcement learning an agent interacts with an environment, gets a reward for correct actions and a penalty for wrong ones, and learns the best strategy by trial and error. Example: a game-playing AI such as AlphaGo, or a self-driving car.

Q7. What is a Neural Network? [2]

Answer & Explanation:

A neural network is a computing model inspired by the neurons of the human brain, with many connected nodes (neurons) arranged in an input layer, hidden layer(s) and output layer. It learns complex patterns from data and is the basis of deep learning.

Q8. What is Deep Learning? [2]

Answer & Explanation:

Deep Learning is an advanced branch of machine learning that uses deep neural networks with many hidden layers to learn very complex patterns from large amounts of data, automatically identifying important features. Examples: face recognition, speech recognition, automatic translation.

Q9. What is Natural Language Processing (NLP)? Give one example. [2]

Answer & Explanation:

NLP is the branch of AI that enables a computer to understand, interpret and generate human language. Example: Google Translate (translation), chatbots, or voice assistants like Alexa/Siri that understand speech and respond.

Q10. What is Computer Vision? [2]

Answer & Explanation:

Computer Vision is the branch of AI that enables a computer to "see" and understand meaningful information from images and video. Examples: face recognition, number-plate detection, X-ray analysis in medicine.

Q11. Mention two applications of AI in finance. [2]

Answer & Explanation:

(1) Fraud detection — identifying abnormal, fraudulent transactions. (2) Algorithmic trading — buying/selling shares automatically. (Extra: risk assessment for loans/credit.)

Q12. What is a chatbot? Where is it used? [2]

Answer & Explanation:

A chatbot is an AI-powered software that automatically converses with people in text or voice and answers their queries. It is used in customer service, online shopping sites, banks and helpdesks to provide 24-hour customer support.

Q13. What is meant by bias in an AI system? [2]

Answer & Explanation:

Bias is the unfair behaviour of an AI system in which it unjustly favours or discriminates against a particular group. It mainly arises from biased training data — for example, a hiring AI that favours one gender over another.

Q14. State two impacts of AI on employment. [2]

Answer & Explanation:

(1) Negative: repetitive, routine jobs become automated, creating a risk of job displacement. (2) Positive: new kinds of jobs and skills are created in AI, data science and robotics.

GROUP – C (Broad / Descriptive Type Questions) [5 × 7 = 35]

Q1. (a) Define Artificial Intelligence and explain its scope. (b) Differentiate AI from human intelligence with examples. [4+3]

Answer & Explanation:

(a) Definition & Scope: AI is the branch of computer science that builds intelligent machines able to think and act like humans. Its scope is broad:

- **Healthcare:** disease diagnosis and treatment support.
- **Finance:** fraud detection and algorithmic trading.
- **Transport:** self-driving cars.
- **Education:** personalized learning and intelligent tutoring.
- **Entertainment & Language:** recommendation systems, NLP, chatbots.

(b) AI vs Human Intelligence:

Human Intelligence	Artificial Intelligence (AI)
Natural, inborn and experience-based.	Man-made, data- and program-driven.
Has emotion, creativity and common sense.	Emotionless, skilled only at specific tasks.
Gets tired, can make mistakes.	Tireless, fast and accurate (if data is correct).

Example: a doctor treats a patient with experience and empathy, while AI quickly analyses thousands of X-rays to help detect disease.

Q2. (a) Explain the three main types of Machine Learning. (b) With a diagram, describe how a neural network works. [4+3]

Answer & Explanation:

(a) Three types of Machine Learning:

- **Supervised Learning:** trained on labelled data; learns the input-output relationship. Example: spam detection.
- **Unsupervised Learning:** finds hidden patterns/clusters in data without labels. Example: customer segmentation.
- **Reinforcement Learning:** learns the best strategy through reward and penalty by trial and error. Example: game-playing AI.

(b) How a Neural Network works: it is arranged in three layers —

$\boxed{\text{Input Layer}} \longrightarrow \boxed{\text{Hidden Layer(s)}} \longrightarrow \boxed{\text{Output Layer}}$

The input layer receives data; every connection has a weight. In the hidden layer, an activation function is applied to the weighted sum (input × weight) to make decisions; the output layer gives the final result. If wrong, the network corrects the weights through backpropagation and gradually learns.

Q3. (a) What is Deep Learning? (b) Explain NLP and Computer Vision with real-life examples. [3+4]

Answer & Explanation:

(a) Deep Learning: Deep learning is an advanced branch of machine learning that uses deep neural networks with many hidden layers to learn very complex patterns from huge amounts of data, automatically identifying important features. Examples: face recognition, speech recognition, automatic translation.

(b) NLP & Computer Vision:

- **NLP (Natural Language Processing):** enables a computer to understand and generate human language. Real examples: Google Translate, chatbots, Alexa/Siri, email spam filters.
- **Computer Vision:** enables a computer to "see" and understand images and video. Real examples: face unlock on phones, road/pedestrian detection in self-driving cars, X-ray/MRI analysis in medicine.

Q4. (a) Describe any three applications of AI in finance. (b) Explain how AI helps in education through personalized learning and intelligent tutoring. [4+3]

Answer & Explanation:

(a) AI in Finance:

- **Fraud Detection:** learns normal transaction patterns and instantly flags fraudulent ones.
- **Algorithmic Trading:** analyses the market and buys/sells shares automatically at the right time.
- **Risk Assessment:** analyses customer data to assess loan or investment risk.

(b) AI in Education:

- **Personalized Learning:** arranges lessons and practice tailored to each student's pace, strengths and weaknesses.
- **Intelligent Tutoring System:** acts like a virtual teacher giving instant feedback, doubt clearing and assessment (e.g., Duolingo, Khan Academy).

Q5. (a) Discuss bias and fairness in AI systems. (b) Explain the impact of AI on employment and social inequality. [3+4]

Answer & Explanation:

(a) Bias & Fairness: Bias is the unfair behaviour of AI, arising mainly from biased or incomplete training data; as a result AI may make unjust decisions against a group (e.g., discrimination in hiring or loans). Ensuring fairness needs diverse and balanced data, transparency, and regular audits.

(b) Impact on Employment & Social Inequality:

- **Employment:** some jobs shrink as routine work is automated, but new AI/data-driven jobs are created — so reskilling is essential.
- **Social Inequality:** without access to advanced technology and education, the gap between rich and poor (the digital divide) can widen.
- **Solution:** training, fair policies and equal access to technology.

Q6. (a) Give a historical overview of AI with key milestones. (b) Explain the role of AI in customer service. **[4+3]**

Answer & Explanation:

(a) Historical overview of AI:

Year	Milestone
1950	Alan Turing proposed the "Turing Test".
1956	John McCarthy coined the term "AI" at the Dartmouth Conference.
1997	IBM Deep Blue defeated Garry Kasparov at chess.
2011	IBM Watson won the quiz show "Jeopardy!".
2022+	Generative AI such as ChatGPT became popular.

(b) AI in Customer Service:

- **Chatbots & Virtual Assistants:** instant, 24×7 answers and support.
- **Automation:** quick solutions to common queries, reducing wait time and cost.
- **Personalization:** analyses a customer's history to give suitable advice and product recommendations.

Q7. (a) Explain the ethical concerns of AI. (b) Suggest measures to make AI systems fair and unbiased. **[3+4]**

Answer & Explanation:

(a) Ethical concerns:

- **Bias & discrimination:** unfair decisions.
- **Privacy:** risk of misuse of personal data.
- **Accountability:** who is responsible for a wrong decision.
- **Employment:** social impact of job losses.

(b) Measures for fair, unbiased AI:

- Use diverse and balanced training data.
- Build transparent and explainable AI.
- Conduct regular audits, testing and human oversight.
- Follow clear ethical guidelines and government regulation.

END OF EXPLAINED ANSWER PAPER

Unit: Foundation of AI — Prepared as per WBCHSE question pattern.